

Notes-Class 10-Science & Technology-2-Chapter-10-Disaster Management-Maharashtra Board

Disaster Management

Class 10-Science & Technology Part-2-Chapter-10- Maharashtra Board

Notes

Disaster : Disaster is the incidence that occurs suddenly causing heavy damage to life and property. The disaster can be man-made or due to natural reason.

OR

Hazardous event that suddenly take place in the surroundings is called disaster.

- **According to the definition of disaster as given by United Nations** : The sudden event that leads to the huge loss of life and property is called disaster.
- **Natural disasters** : Floods, wet and dry famine, cyclones, earthquakes, volcanoes, etc. are caused due to sudden change in the environment causing great damage. Since these disasters are due to nature's wrath, they are called natural disasters.
- **Man-made disasters**: When mankind needs natural resources for developmental purpose, environment gets damaged. Such disasters are due to man-made actions and thus they are called man-made disasters.

Many **natural disasters** and related unprecedented situations have been occurred in the Asian continent and region of Pacific Ocean. Huge loss to the life and the planet Earth has been occurred due to such disasters.

Increasing population, its increasing needs emerging problems out of it are now at the extreme end. Such disasters have been increased after the world war-II. Condition of instability arises in the country due to various reasons like economic inequality, racial and religious differences, etc. Incidences like terrorism, abduction, social differences have been a routine now.

Production and use of harmful chemicals is under ban in developed countries. However, production of either same or those chemicals which can wipe out the human race is common in developing and poor countries.

Atomic energy plants are used for generation of nuclear energy. But they emit hazardous radiations. The accidental leakage in atomic energy plant that occurred at Chernobyl, Russia is still producing ill-effects. Nowadays, many countries are equipped with atomic energy. At the same time risk of radiation leakage and allied accidents have also increased.

Classification of disasters :

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Disasters can be classified depending on following main criteria.

- The extent of the damage caused.
- The period of disaster.
- The long term effects.
- The reason for the disaster.

There are two main types of disasters, natural and man-made. This is the **preliminary Classification of disasters**.

Second method of Classifying disasters :

- **Catastrophic disasters** - Cyclones in Odisha, catastrophic earthquakes of Gujarat and Latur, frequently buzzing cyclones in coastal Andhra Pradesh, etc.
- **Disasters making the impact for long duration** : Famine, various problems of crops, strikes of workers, rising levels of oceans, desertification etc.

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Types of disasters :

Types of disasters : There are three types of disasters, viz. geophysical, biological and man-made.

Geophysical :

- **Geological** : Earthquake, volcano, tsunami, landslides, landfall, erosion, alkalization, flooding, etc.

- **Climatic** : E.g. Hot and cold waves, snow storms, snowfall, cyclones, hailstorm, drought, flood, meteorite, sunspots, etc.

Biological :

- **Animal-origin** : Infectious viruses, bacteria (cholera, malaria, hepatitis, plague, bird flew), insects, bite of venomous animals, etc.
- **Plant-origin** : E.g. Forest fire, fungal disease spreading (Blister), weed, (aquatic, carrot grass, common grass)

Man-made:

- **Accidental or Unintentional** : Toxic gases, Atomic test, Unplanned action, Accident
- **Intentional** : War, fire, bomb blast, forced migration, terrorism, rapes, child labour.

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Effects of disasters :

- The great loss to life and property of the people.
- Different areas of life like economic, social, cultural, political, law and administration, etc. are affected due to sudden disasters.
- The life in the affected area is totally disturbed.
- Nature and duration of each disaster is different. The period of disaster decides whether it is short term or long-term.
- Disasters are caused due to different reasons. Each disaster has its own environmental impact.

Actual effects :

- Effects of every type of disasters are different.
- **Floods** : Collapsing of bridges, flooding of coastal villages, shortage of food .
- **Earthquake** : Collapsing of houses, developing cracks in land.
- **Forest fire and drought** : Adverse effect on the environment.

Effects on the economy of the nation and local leadership :

- Huge expenses on the reconstruction of the destroyed structures.
- Long term effect on the economy.
- If local leadership is weak, the victimized citizens become confused.
- During disasters administrative problems arise.
- Local governing bodies too are affected.
- The Government departments also are unable to solve the problematic situations caused by the disaster.
- Entire system collapses as all the concerned departments are affected by disasters.

Effects of Heavy Rain :

Effects of Heavy Rain :

Due to heavy rain following effects are observed in surrounding area,

- There was waterlogging in all the shallow areas.
- All the transport systems get collapsed on that day.
- Large trees fell down injuring the people.
- The water logging caused condition like floods.
- Schools, colleges and offices were shut down.
- People were caught in troublesome situation.

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Destructive effects of flood:

Destructive effects of flood:

- The flood waters uproot the trees, the houses collapse due to surge of water.
- The fields get waterlogged. The water does not recede.
- The cattle die of asphyxiation by drowning.
- Since everything is under water, the process of decomposition begins at a higher rate. This may spread the epidemics of the diseases.

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Effects of dry famine :

Effects of dry famine :

- In dry famine, there is scarcity of water.
- The crops die as there is no proper irrigation.
- There is severe food shortage.
- Due to dearth of water, the cattle die and human beings are forced to migrate.

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Effects of earthquake :

Effects of earthquake :

- The earthquake is responsible for large scale devastation.
- The houses collapse. The roads are damaged making the transportation impossible.
- The electricity and water supply is disrupted due to damage.
- There is large scale damage to lives and property.

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Effects of forest fire on environment :

Effects forest fire on environment : The largely spread fire of the dry grass, shrubs and trees in the forest due to heightened temperatures is called forest fire.

- The effect of forest fire on the environment is greatly devastating.
- Biodiversity is lost. Plants and animals die due to fire. T
- The vegetation becomes dead as the plants turn into ash.
- The atmosphere is full of smoke and hence causes air pollution.

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Nature and scope of disaster: Some of the important aspects of the disaster become clear only when the nature and scope of the disaster is understood.

For this following aspects have to be studied thoroughly :

- Exact nature of the disasters.
- The changes in the nature before the onset of the disaster.
- Time period of impact of disaster.

The scope of the disaster is dependent on the following facts :

- Pre-disaster phase
- Warning phase
- Emergency phase
- Rehabilitation phase
- Recovery phase
- Reconstruction phase

Three sensitive phases of the disaster :

Three sensitive phases of the disaster :

1. Phase of emergency

- Maximum lives can be saved by quick actions in this phase.
- Actions such as search and rescue operations, medical assistance, first aid, restoring communication services, removing the people from affected area, etc. are done in this phase.
- Estimating the gravity of disaster is done in this phase only.

2. Transitional Phase

- In this phase the rehabilitation work starts.
- Clearing of debris, restoring water supply, repairing roads, etc. is done.
- Bringing normalcy in public life.
- Different institutes offer the monetary/ other help. This help is distributed to victims.
- Victims are offered with the permanent means of earning livelihood.
- Soothing the mental stress of victims.
- Rehabilitation of the victims is important aspect of this phase.

3. Reconstruction Phase

- Highly complicated phase.
- Begins in transitional phase. Reconstruction of buildings and roads and water supply facilities are restored.
- Restarting of farming practices.
- This phase takes long time for reconstruction.

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Disaster management :

- Disaster management is the essential action that has to be undertaken by each nation and its citizens.
- Disaster management schemes should be dependent on every location, time and nature of the disaster.

The disasters are inevitable but we have to overcome the hazardous after-effects of these. This fact has led to concept of disaster management.

- Effective disaster management is necessary in any kind of disaster. It may be minor or major, spread over long period of time or it may be short term.
- The disaster management should involve public participation. Disaster hits in a rapid way.
- In managing the after-effects of disasters, there should be arrangements to cope up with disastrous effects or there should be ways to make citizens capable to face and fight against the consequences.
- The losses should be minimized. Disasters are suddenly forcing us to deal with them but by disaster management techniques, the losses can be minimized.

Objectives of disaster management :

Objectives of disaster management :

- Saving human life from disasters. Helping them to move away from the place of disasters.
- Essential commodities to be supplied to the affected people. This will help to reduce the gravity of disaster.
- To bring back the conditions to normalcy.
- Rehabilitation of the affected and displaced victims.
- Protective measures for future in order to 'develop capability to face the disasters in future.

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Disaster management planning :

- Through scientific and careful observations and analysis of previous data, the suitable action plans are developed for the disasters that might come in future.
- In such plans the preventive measures, rehabilitation and reconstruction is included. Such plans are kept ready for future.
- Disaster management is divided into pre-disaster management and post-disaster management.

Pre-disaster management :

Pre-disaster management : Before the disaster strikes, there are some measures taken to rescue people or to keep preparedness for the future calamities. This is called pre-disaster management.

It includes following points :

- To identify the disaster prone areas.
- Through predictive intensity maps and hazard maps, the information should be collected about the probability of future disaster's intensity and probable sites of disasters, respectively.
- Learning by ourselves and then imparting special training to people to tackle with future disasters.
- Making general public aware about disaster management by training them through programmes. Spreading the message through mass media, etc.

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Post-disaster management :

Post-disaster management :

- Helping the victims of disasters in all possible ways.
- Local people who are saved from disaster should participate for further rescue work.
- Establishing the help centres for victims. Controlling centres, established are according to the type of disaster.
- Materials provided from the control centres for the victims are categorized, and distributed the same to victims with continuous review of the help.

- Preparedness for the rescue operations.

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Main aspects of disaster management cycle :

Main aspects of disaster management cycle :

- **Preparation** : If there is disaster in future, then a plan for minimizing the destruction is kept ready. This is called preparation for the disaster.
- **Preparedness** : In case of disaster, the general public and administration should immediately rise for action. This action is planned. This aspect is called preparedness.
- **Impact of disaster** : After the disaster recedes, its intensity is studied. How it has impacted is also analysed along with disaster management.
- **Response** : Just before the disaster and immediately after the disaster, the response given is to be analysed in this step.
- **Resurgence** : Resurgence forms an important link between emergency measures and national progress. It is a step showing extent of welfare and rehabilitation work done after the disaster. The progress of nation also becomes clear after analysing resurgence.
- **Redemption** : The damage is caused to the people of the nation, thus the planning is done to minimize such damage.
- **Restoration** : The return of the normalcy after the disaster is called restoration. The restoration time period for coming back to normal routine after the disaster may be different for different types of disasters.

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Natural disasters cannot be avoided. But by proper management, the loss can be minimized. Man-made disasters can be avoided if care is taken in our actions and behaviour. In situations of disasters we should help each other. This is the ethical responsibility of all human beings.

Functions of Disaster Management Authority :

Functions of Disaster Management Authority :

- Giving the necessary instructions.
- Assessing whether the instructions are followed or not.
- Reviewing the entire system.
- Controlling the actions.
- Planning and executing the plans.
- Designing schemes for each district, separately for each type of disaster and getting those sanctioned from state-level authorities.

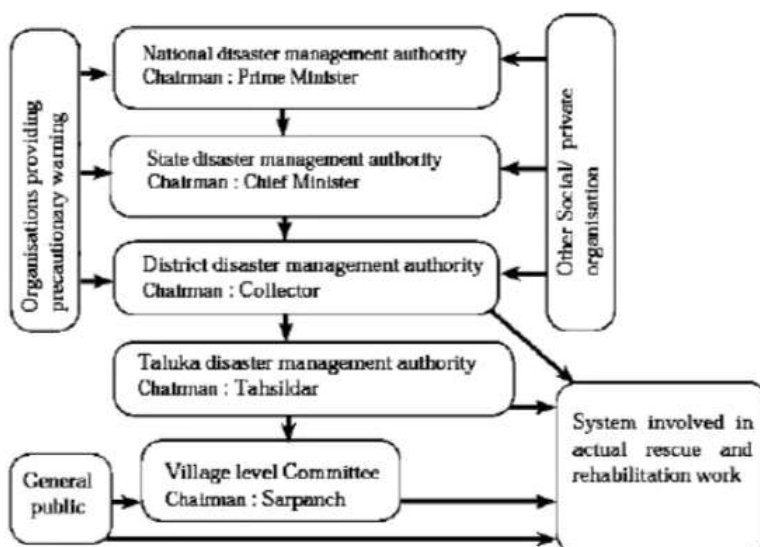
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Structure of Disaster Management Authority :

Structure of Disaster Management Authority :

The Government has established an authority for management of disasters. Indian Government has also passed Disaster Management Act, 2005.

Control and coordination of disaster management is carried out from national level right up to village as shown in the following chart :



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District Disaster Management Authority : District collector is responsible for disaster management and implementation of rehabilitation schemes for that district. Planning, coordinating and controlling the implementation of rehabilitation programme is done by District Collector.

District-wise Disaster Control Unit :

- Establishment of district control unit is done immediately either after the impact of disaster or due to the warning of forthcoming disasters.
- Help and guidance from various agencies like army, air force, navy, telecommunication department, paramilitary forces, etc. is taken to review the various aspects of disaster.
- The control unit also coordinates with various voluntary organizations that help in disaster management.

The international organizations :

The international organizations that work in field of disaster management :

- United Nations Disaster Relief Organization.
- United Nations Centre for Human Settlements.
- Asian Disaster Reduction Centre.
- Asian Disaster Preparedness Centre.
- World Health Organization.
- United Nations Educational, Scientific and Cultural Organization.

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National Disaster Response Force :

National Disaster Response Force :

- The army jawans constitute the National Disaster Response force.
- It was established as per the Disaster Management Act, 2005.
- 12 divisions of this force work in the country.
- Its headquarter is located at Delhi and the army takes any required action throughout all the states of India.
- In Maharashtra, State Reserve Police Force carries out responsibilities of National Disaster Response Force.
- Jawans of this force help in rescue work at the times of disasters like cyclones, cliff-sliding, building collapse, etc.
- Website : <http://www.ndrf.gov.in> gives the information of National Disaster Response Force.

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First Aid and Emergency Action : When suddenly any disaster takes place, the help given to the victims before the medical treatment is available, is called first aid. This knowledge of first aid becomes useful to save one's life.

- In day-to-day life, we come across many such disastrous incidences where such help is needed. Varieties of disasters like accidents, stampede, injuries in fighting, electric shock, burns, heat shock, snake bite, dog bite, fire due to electric short circuit, epidemic of any disease, etc. are the disasters in which first aid becomes essential.
- In disasters, the victim's condition is to be taken into account before starting any first aid. As per the condition, various **transportation methods** like cradle method, carrying on back, carrying on two hands are to be followed.
- Sometimes we face the disaster due to our own unawareness. We have to watch out for the **symbols** that warn us against the potential dangers. Such

First Aid Kit :

First Aid Kit : Material necessary for first aid is kept in the first aid kit. Such first aid kit should be available at every place such as in schools, colleges, malls, company and offices, etc..

The necessary material in first aid box:

- For wrapping the wounds : Bandage strips of different size, Wound gauze, Triangular and circular bandage, Medicated cotton, Band aid
- For cleanliness; Hand gloves, Clean and dry cloth pieces, Soap
- Medicines : Antiseptic (Dettol/ Savlon), Petroleum jelly
- Safety pins, Blade, Small pins, Needle, Scissors
- Thermometer, Torch.

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Mock Drill: Mock drill is the activity which is carried out by creating virtual or apparent situation of disaster.

- It is carried out to check the preparedness if the disaster approaches.
- The responses of the people, the quickness of actions, etc. can be analysed prior to the actual disaster happens.

- Trained personnel check execution of plan designed for disaster redressal. They are responsible for the training of people in case of disasters. The whole exercise helps to check the efficacy of the system prepared for disaster redressal.
- Mock drill is organized in various schools to create the awareness among the school children in case of disasters. Such drills are taken with the help of fire fighters, by police force and by some voluntary organizations.

Objectives of Mock Drill :

Objectives of Mock Drill :

- Evaluating the response of the common people to the disaster.
- Improving the coordination between various departments that work for the disaster control.
- Identifying one's own abilities at the time of disaster.
- Developing the ability to respond quickly to disaster.
- Checking the competency of the planned actions.
- Assessing the possible errors and risks.

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Important notes for our safety :

Important notes for our safety :

- Not to crowd and push each other while using staircase in the school.
- Important help-lines should be remembered and used whenever need arises: Police 100, Fire fighting force 101, Disaster Control Unit 108, Ambulance 102, Child helpline : 1028 etc.
- Always follow the traffic rules. Looking first at left and then at right sides while crossing the road.
- Not to touch any unclaimed object.
- Not to spread rumours.
- Not to cause chaos at the crowded places.

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